Triadic Casimir — PKLSA v0.3.4 Framework Report

SST Workbench / Unregistered instance

October 19, 2026

1 Instance identity

A065-v0.3.4 versioned Triadic-Casimir research instance.

2 Research question

Does a preregistered triadic clock/stress geometry pilot satisfy its analytic controls while independent physical coupling gates remain unresolved?

3 Null and alternative hypotheses

No independently qualified source-specific physical two-surface response is established by the present geometry and toy models.

4 Explicit assumptions

Time-dependent regularized filament RK4 is exploratory and is not full Euler closure.

5 Definitions

u material velocity; omega=curl u; S symmetric strain.

6 Analytical equations

The official AUTO_SCIENCE_CONTRACT fragment records frozen equations.

Machine-readable science contract

Research question Does a preregistered triadic clock/stress geometry pilot satisfy its analytic controls while independent physical coupling gates remain unresolved?

Objective Independently verify three dataset seals, analytic Euler/gauge/sheet controls and one sourced KnotPlot trefoil finite-core geometric response without importing target force.

H_0 No independently qualified source-specific physical two-surface response is established by the present geometry and toy models.

H_1 Independently qualified PKLSA Euler motion, boundary conditions and microscopic link coupling produce held-out physical predictions.

Registered equations

E1: Incompressible Euler kinematics

$$\nabla \cdot \mathbf{u} = 0, \quad \boldsymbol{\omega} = \nabla \times \mathbf{u} \tag{1}$$

Dimensional check: divergence and vorticity both inverse seconds

E2: Pressure Poisson source

$$\nabla^2 p = \rho_M(\frac{1}{2}|\omega|^2 - ||S||_F^2) \tag{2}$$

Dimensional check: kg/(mˆ3 sˆ2) on each side

E3: Regularized geometric velocity diagnostic

$$\mathbf{u}(\mathbf{x}) = \frac{1}{4\pi}\sum_j \frac{\Delta\mathbf{X}_j \times (\mathbf{x} - \mathbf{X}_{j+1/2})}{(|\mathbf{x} - \mathbf{X}_{j+1/2}|^2 + \varepsilon^2)^{3/2}} \tag{3}$$

Dimensional check: dimensionless Gamma=1 geometry diagnostic

Registered code-to-mathematics steps

1. **S1** [G1; equations E1]: Verify all three source seals before any scientific result is evaluated
2. **S2** [G2; equations E2, E3]: Evaluate analytic Euler controls and preregistered trefoil geometry resolution scan
3. **S3** [G3; equations E3]: Withhold physical Casimir claims without independent providers, wall dynamics, and atom-link normalization
4. **S7** [G12; equations E3]: Audit historical Fourier and torus XYZ hash and analytic geometry
5. **S8** [G13; equations E3]: Compare reduced vortex filament RK4 spatial/time resolution on independent geometric seeds
6. **S9** [G14; equations E3]: Report fixed core regularization sensitivity without physical force interpretation
7. **S4** [G10; equations E3]: Verify fixed-core 150/300/600 convergence, separately retaining original G2 failures
8. **S6** [G3; equations E1, E2, E3]: Evaluate G3 only after precommitted G1+G10 and temporal/analytic guard; retain G2=FAIL and physical UNRESOLVED
9. **S5** [G11; equations E3]: Verify exact Fremlin raw hash and native C++ time-series refinement; never infer real Euler solution

7 Scientific algorithm

Verify three seals; compute controls; evaluate trefoil diagnostic.

8 Dataset provenance

D01 D02 D03, KnotPlot N300, external Fremlin and separate 2015 historical Fourier and torus geometric seeds. No historical geometry is physical PKLSA evidence.

9 Independent gates

G12/G13/G14 are historical geometry and numerical diagnostics only, independent of G3. No physical conclusions.

10 Numerical controls

Dimensionless finite-segment Biot-Savart and finite-difference derivatives.

11 Backend authority

CPU profile enables strict C++ gate; G3 may be evaluated but physical validation remains unresolved.

12 Blindness and integrity

All three dataset SHA256 seals checked; reveal comparator not used in blind kernel.

13 Statistics

Geometry subsamples are not independent upstream providers.

14 Result status

Read dynamic AUTO_RESULTS fragment; no target-matched force.

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	94be38e2e2d6f3226c408e5a8a7ce53bc5b400373da4796ed2354e6ec2f8f0d6
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	52bcd25f823f0a47ac03671d51c195f492580b3b8d3e4f17aba95611a2e40dd3

Gate ledger

Gate	Status	Question
G0	PASS	Is the frozen protocol intact and is the runtime self-consistent?
G1	PASS	Are all preregistered sources present, identified and provenance-qualified?
G2	FAIL	Do reference numerics satisfy analytic/sanity and dimensional checks?
G10	PASS	Does refined KnotPlot spatial RK4 filament discretization pass 150/300 and 300/60
G11	FAIL	Does authentic Fremlin C++17 provider satisfy source, spatial and time checks (sin
G12	PASS	Are historical Fourier and torus XYZ inputs hash-valid and analytically consistent?
G13	FAIL	Do both historical geometries satisfy preregistered reduced filament ODE resolution
G14	PASS	Were epsilon sensitivity diagnostics recorded without fitting physical claims?
G5	PASS	Does actual C++17 FP64 filament velocity match Python at 75/150/300? Kernel-o
G3	UNRESOLVED	Does target-free discovery reject the preregistered null under the registered statistic
G4	NOT_RUN_PREREQUISITE	Does an independent held-out/provenance-family confirmation reproduce the discov
G6	NOT_APPLICABLE	If enabled, does GPU screening pass CPU to GPU parity without being promoted b
G7	NOT_APPLICABLE	Does the result replicate across preregistered source families/encodings without pseu
G8	NOT_RUN_PREREQUISITE	Are physical PKLSA Euler, material, surface and independent-provider closure requ
G9	DEFERRED	After all blind gates, does reveal verification match the preregistered commitment a

Backend manifest


```
{
  "native_kernel": {
    "authority": "KERNEL_ONLY_NOT_FRAMEWORK_G5",
    "backend_executed": "cpp17-serial-fp64",
    "comparisons": 1575,
    "compiler": "g++",
    "expected_component_count": 1575,
    "max_abs_component_error": 0.0,
    "max_relative_component_error": 0.0,
    "meaning": "Velocity-kernel parity only. NOT full RK4, Euler PDE, physical G5 or PKLSA dynamics.",
    "requested_backend": "cpp-17-binary64",
    "rms_component_error": 0.0,
    "schema": "A065-CPP-KERNEL-QUALIFICATION-1",
    "source_sha256": "11bec24153e8471f94e43dfeb0bae7b675fcb84a240e7d72092d08da7528fabd",
    "status": "PASS",
```

```
    "threshold": 5e-11
  },
  "physical_force_status": "UNRESOLVED",
  "pklsa_geometry": "a60b1a95a00110e3cb32087e7b4328d16e967379ebf97d60117b0ec05692ffb2",
  "provider_manifest_sha256": "74cc9349324a15f4c1d4ca1a6910845af45aff69512538ec362ea4ae674383dc",
  "provider_status": "PASS_RAW_HASH_AND_XYZ_PARSER_ONLY",
  "python_reference": {
    "actual_backend": "python-stdlib-binary64",
    "authority": "REFERENCE",
    "precision": "float64"
  },
  "schema": "SST-BACKEND-MANIFEST-2",
  "scope": "Kernel-only native gate; complete coupled physical validation withheld",
  "sealed_source_groups": [
    "D01_EULER",
    "D02_CLOCK_LINK",
    "D03_TWO_SURFACE"
  ],
  "v033_numerical_routing": {
    "expected_failure_ids": [
      "D01_G08_FILAMENT_SPATIAL_REFINEMENT",
      "D01_G10_FILAMENT_SPATIAL_75_300"
    ],
    "g10_status": "PASS",
    "g3_permitted_outcome": "UNRESOLVED",
    "historical_G2": "FAIL_RETAINED",
    "observed_failure_ids": [
      "D01_G08_FILAMENT_SPATIAL_REFINEMENT",
      "D01_G10_FILAMENT_SPATIAL_75_300"
    ],
    "physical_force_status": "UNRESOLVED",
    "prereg_sha256": "3856a587779d82df3b11ad1e64c4e4a45b6b4318e30c00676e4015d4df841fd6",
    "protected_mismatches": [],
    "ready_for_G3_evaluation": true,
    "reference_count": 19,
    "release": "v0.3.3",
    "schema": "A065-ROUTING-DECISION-V033-1",
    "scope": "Execution-eligibility diagnostics only. NOT Casimir, gravity, full Euler or PKLSA physical
    "time_discretization": "PASS"
  },
  "v034_historical": {
    "gate_scope": "GEOMETRY_SIMULATION_ONLY",
    "native_status": "EXECUTED",
    "prereg_sha256": "73545eaddc62a3f2eda0553610220812758dc628ad2aa0eb8d4e9177d36207e5"
  }
}
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Source manifest


```
{
  "manifest_sha256": "2056c878e3384905d5ea1c37a34acefa39b2e26a127272cca00946b5eb78b3f6",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "A046",
      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "synthetic-euler",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Casimir_B
      "provenance_family": "synthetic-euler",
      "registry_hint": {
        "catalog_id": "A046",
```

```
        "hits": []
      },
      "role": "analytic_control",
      "sha256": "9c4da362ab58a546fe8aff349a024dcc7fb63218ef30053dceee3d6659c574ad",
      "source_id": "D01",
      "version": "v0.3.4"
    },
    {
      "catalog_id": "E010",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "knotplot-relaxed-3_1",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Casimir_B
      "provenance_family": "knotplot-relaxed-3_1",
      "registry_hint": {
        "catalog_id": "E010",
        "hits": []
      },
      "role": "geometry_diagnostic",
      "sha256": "a60b1a95a00110e3cb32087e7b4328d16e967379ebf97d60117b0ec05692ffb2",
      "source_id": "PKLSA",
      "version": "v0.3.4"
    },
    {
      "catalog_id": "CANON",
      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "synthetic-link",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Casimir_B
      "provenance_family": "synthetic-link",
      "registry_hint": {
        "catalog_id": "CANON",
        "hits": []
      },
      "role": "toy_control",
      "sha256": "8e4bcfcdf186e539f0cdeafd9cbbf9f31dbe7bbfb95c14d89099e6c31f760af8",
      "source_id": "D02",
      "version": "v0.3.4"
    },
    {
      "catalog_id": "CANON",
      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "synthetic-sheet",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Casimir_B
      "provenance_family": "synthetic-sheet",
      "registry_hint": {
        "catalog_id": "CANON",
        "hits": []
      },
      "role": "heldout_control",
      "sha256": "fff8860c2597840d0f15906a43f8710793c0909ebfdde1b1eb6f64aea7863adc",
      "source_id": "D03",
      "version": "v0.3.4"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Cas
}
```

15 Scientific interpretation

Physical Casimir force and gravitation remain unvalidated.

16 Reproduction

Run native FREEZE and FULL via the installation bridge.